CHAPTER 1 - Web Application Basics

1.1 Objectives

The objectives of the course are the below

- To understand the client server architecture
- To know the process of sending requests and get responses from a server
- To understand the types of servers (Web Servers, Application Servers)
- To understand the differences between Web Servers and Application Servers
- To understand deployments and the knowledge about jar and war files

1.2 Audience

This document is designed for Java EE Learners and programmers with a need to understand the basics of Web Applications. The trainees who have understood the basics of Java technology and programmed Core Java programs are the intended audience.

1.3 Prerequisites

Associates should have gone through all the concepts of the core skills phase, specifically the core java concepts.

1.4 Course Content

1.4.1 Introduction

A Web Application in simple terms indicates any software designed to be running or rendered on a web browser. These applications are created using browser supported languages such as Javascript, HTML, Java Server Pages (JSP) and CSS. In simple cases, the application becomes a client that interacts with end users and receives requests from them. These requests are processed and results are sent back accordingly. The software or applications that fulfill the requests sent from a client program are termed as servers.

1.4.2 Client Server Architecture

Client server architecture is used in applications that belong to two, three or n tier categories of applications.

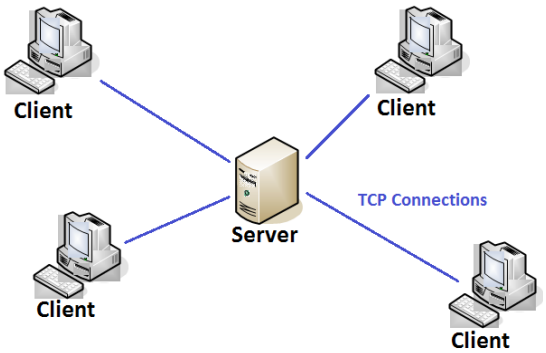

Figure 1: Client Server Architecture

In the image above we have three clients requesting dishes from one server. This is one such comparison between the actual client server with a real life server scenario.

The client server characteristics describe the co-operating nature between software programs within an application. These programs may either be deployed or made available on a single hardware environment or multiple hardware environments. Thus arises the concept of tiers in an application.

The client server architecture in detailed level would involve several concepts like the design of the application in terms of tiers. This facilitates the deployment of each tier on individual environment as a tier application.

Since we deal with the Java EE technologies we will look into three or N tier architecture. The n-tier or multi tier architecture in software application development gives the flexibility of developing modular and reusable applications. By this, maintenance of each module such as presentation, business logic and data layers, is easily possible.

1.4.3 Request and Response cycles

Any web application would deal with content that would be displayed on a browser. Browsers support HTML content to be displayed.

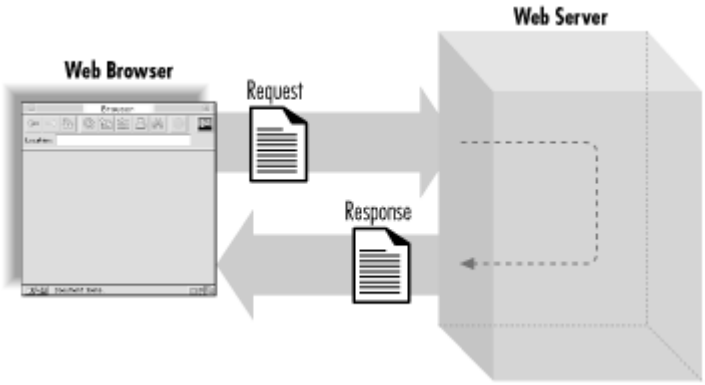

Figure 2: Request and Response Cycle

Every time a user types a URL in the address bar of the browser and expects a page to be displayed that initiation would be considered as a request. Then the server (where the application is deployed and the HTML content is available to the users) finds the requested content and responds to the request otherwise called as response.

1.4.4 Web Servers

A Web server supports Web-based applications.

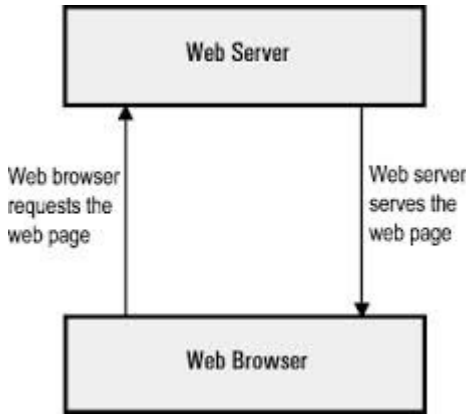

Figure 3: Web Server

A Web server handles the HTTP protocol. When the Web server receives an HTTP request, it responds with an HTTP response, such as sending back an HTML page. To process a request, a Web server may respond with a static HTML page or image, send a redirect, or delegate the dynamic response generation to some other program such as CGI scripts, JSPs (JavaServer Pages), servlets, ASPs (Active Server Pages), server-side JavaScripts, or some other server-side technology. Whatever their purpose, such server-side programs generate a response, most often in HTML, for viewing in a Web browser.

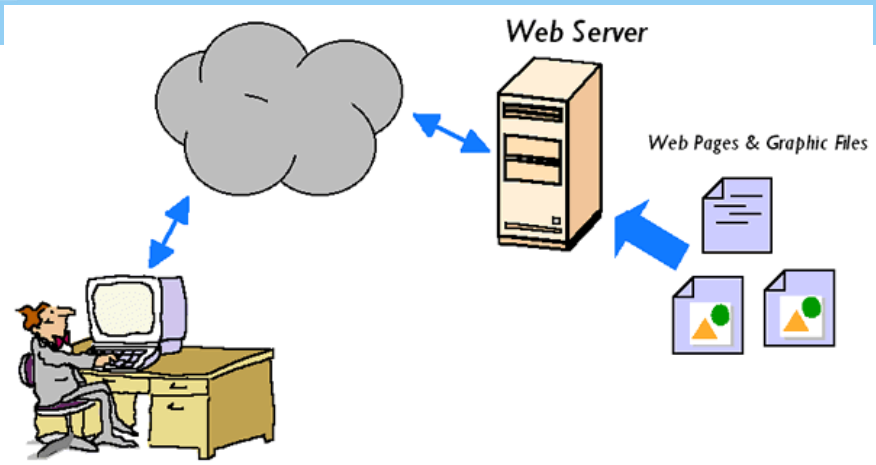

Figure 4: Web Server (Detailed)

Understand that a Web server's delegation model is fairly simple. When a request comes into the Web server, the Web server simply passes the request to the program best able to handle it. The Web server doesn't provide any functionality beyond simply providing an environment in which the server-side program can execute and pass back the generated responses. The server-side program usually provides for itself such functions as transaction processing, database connectivity, and messaging.

While a Web server may not itself support transactions or database connection pooling, it may employ various strategies for fault tolerance and scalability such as load balancing, caching, and clustering—features oftentimes erroneously assigned as features reserved only for application servers.

1.4.4 Application Servers

An application server supports enterprise applications that require interoperability and interaction between several other applications.

The application server exposes business logic to client applications through various protocols, possibly including HTTP. While a Web server mainly deals with sending HTML for display in a Web browser, an application server provides access to business logic for use by client application

programs. The application program can use this logic just as it would call a method on an object (or a function in the procedural world).

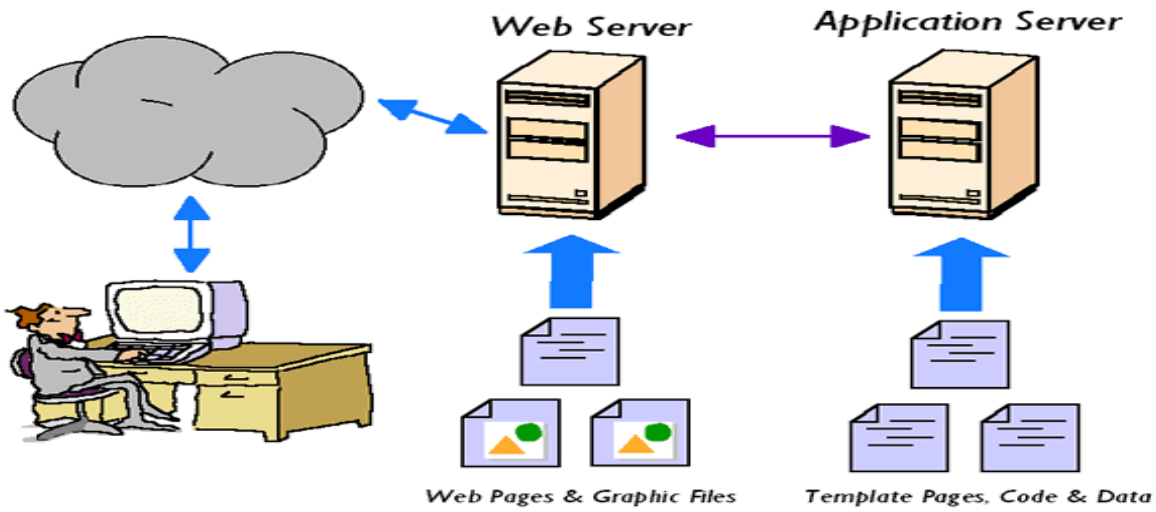

Figure 5: Application Server

Such application server clients can include GUIs (graphical user interface) running on a PC, a Web server, or even other application servers. The information traveling back and forth between an application server and its client is not restricted to simple display markup. Instead, the information is program logic. Since the logic takes the form of data and method calls and not static HTML, the client can employ the exposed business logic however it wants.

In most cases, the server exposes this business logic through a component API, such as the EJB (Enterprise JavaBean) component model found on J2EE (Java 2 Platform, Enterprise Edition) application servers. Moreover, the application server manages its own resources. Such gate-keeping duties include security, transaction processing, resource pooling, and messaging. Like a Web server, an application server may also employ various scalability and fault-tolerance techniques.

1.4.5 Difference between Web and Application Servers

	Application Server	Web Server
What is it?	A server that exposes business logic to client applications through various protocols including HTTP.	A server that handles HTTP protocol.
Job	Application server is used to serve web based applications and enterprise based applications (i.e Servlets, JSP and EJB). Application servers may contain a web server internally.	Web server is used to serve web based applications.(i.e Servlets and JSP)
Functions	To deliver various applications to another device, it allows everyone in the network to run software off of the same machine.	Keeping HTML, PHP, ASP etc files available for the web browsers to view when a user accesses the site on the web, handles HTTP requests from clients.
Supports	Distributed transaction and EJB	Servlets and JSP
Resource utilization	High	Low

1.4.6 Deployments

1.4.6.1 Project Structure

A web application can be deployed by placing the Project having the folder structure as mentioned. The details of the project for a web application would be dealt in the forthcoming chapter.

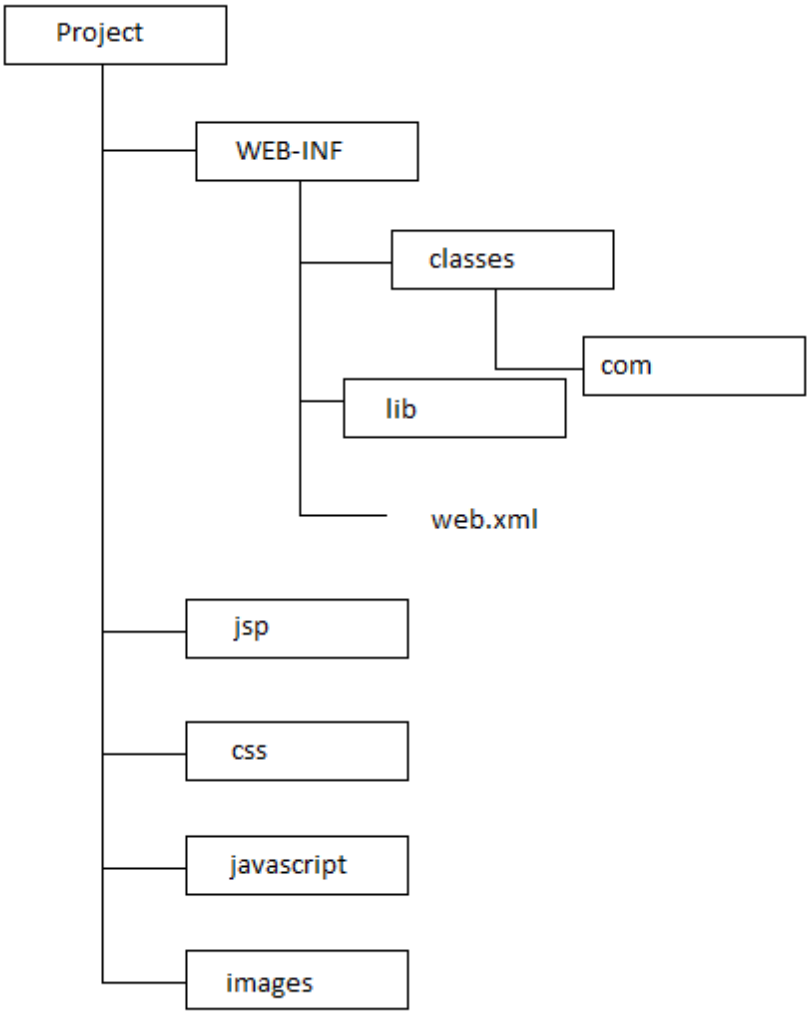

Figure 6: Project structure for Deployment

The project folder can be placed under the webapps folder of the server (either the application server or the web server) directory either as a folder or as a WAR file.

Once the server is started, the application can be invoked using the browser URL like

http://localhost:8080/<ProjectName>/<jspname.jsp>

(The illustrated URL is a sample one for an application running on Tomcat server.

In the above URL the http stands for the use of the HTTP protocol by the browser to transmit HTML content to the hosting server. In this case we use localhost if we're trying to access the local machine. It would work if we replace the word with the IP of your machine as well.

http://172.24.138.415:8080/<ProjectName>/<jspname.jsp>

1.4.6.2 Java Archive (JAR) Files

A JAR (Java ARchive) file is a file that contains the class, image, and sound files for a Java application or applet gathered into a single file and possibly compressed. When a programmer gets a Java program development kit, a small program or utility called "jar" is included. The jar utility lets the programmer create, list, or extract the individual files from a JAR file. In an enterprise, a Java application can be started with a set of JAR files for use during execution. An off-the-shelf open source package can be delivered as a JAR file and run with XML data.

On the Web, a JAR file containing an applet may accompany a Web page. By putting the applet components in a single file and compressing that file, download time is saved. Ordinarily, a browser user will not need to "open" or view a JAR file directly. It is opened when the Web page is received and the applet is in some manner initiated.

The JAR format is based on the popular zip file format. The jar library files used by the project can be placed in the WEB-INF/lib folder along with the other resource jar files.

1.4.6.3 Web Archive (WAR) Files

You can also deploy using WAR files instead of regular directories in deployments. A WAR file is just a JAR file (like a ZIP file) with a *.war* extension. You can create WAR files using jar, WinZip, or the Windows XP "Create Compressed Folder" R-Mouse option. If you use WAR files, a directory such as *myWebApp* should become *myWebApp.war*, and the top-level directory within the WAR file should be *WEB-INF* (i.e., do not repeat *myWebApp* within the WAR file).

Note: For details on how to create a basic web application, refer the support document 1_Content_WebApplication Basics_SupportDoc.pdf

1.4.7 Summary

In this topic, you learnt about the below

- ✓ Client – Server Architecture is used in all the web applications over the network
- ✓ Request and Response Cycle in a browser is used to maintain the life of objects/data over every transaction with the server.
- ✓ Web Server is used to deploy Servlet based applications/Simple web based applications in the Servlet Container
- ✓ Application Server is used to deploy EJB applications and Servlet based applications becoming a enterprise application server.
- ✓ Deployment of a J2EE application requires a standard folder structure.
- ✓ Jar files are used to compress and collate related files and provide them as a package to the developer as APIs
- ✓ J2ee web projects can be boxed into a WAR file and deployed to a webserver by placing the WAR in the webapps folder of the server.